INCH-POUND
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(See 6.8)

MILITARY SPECIFICATION

LIFE PRESERVER, CO_2 INFLATABLE, UNDERWATER DEMOLITION TEAM

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers life preservers of coated nylon fabric construction, CO_2 inflatable type, for use as emergency equipment by underwater demolition teams.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-295 - Thread, Nylon.
QQ-B-613 - Brass Leaded and Nonleaded; Flat Products (Plate, Bar, Sheet, and Strip).
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting.
QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-495 - Finish, Chemical, Black, for Copper Alloys.
DOD-E-699 - Enamel, Exterior, Deck, Gray (Formula No. 20). (Metric)
MIL-R-900 - Rubber Gasket Material, 45 Durometer Hardness.
MIL-L-3454 - Life Preservers, Packaging of.
MIL-P-5516 - Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160°F.
MIL-A-5540 - Adhesive, Polychloroprene.
MIL-L-15581 - Life Preservers, CO_2 Inflatable, Pouch Type.
MIL-C-16385 - Cartridge, Carbon Dioxide, Inflatable Life Preserver Use.
MIL-W-17337 - Webbing, Textile, Woven Nylon.
MIL-C-19002 - Cloth, Coated, and Strip, Coated Cloth - Polychloroprene on Nylon.
MIL-C-19377 - Cloth, Twill, Nylon.
MIL-C-25369 - Cylinders, Carbon Dioxide, for Inflation of Pneumatic Life Preservers.
MIL-O-81375 - Oral Inflation Assemblies, Survival Equipment, Inflatable.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

AIR FORCE - NAVY AERONAUTICAL

AN-227 - Fasteners, Snap and Curtain.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

S3306-1183395 - Rubber Sleeve Check Valve for CO_2 Inflatable Life Preserver.
S3306-1220372 - Life Preserver, CO_2 Inflatable, Underwater Demolition Team Assembly.
S3306-1220373 - Life Preserver, CO_2 Inflatable, Underwater Demolition Team Subassembly.
S3306-1220374 - Life Preserver, CO_2 Inflatable, Underwater Demolition Team, Details.
S3306-1220375 - Life Preserver, CO_2 Inflatable, Underwater Demolition Team: Harness Assembly and Details.
S9300-1411524 - CO_2 Holder and Holder Gage.

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(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 413 - Standard Test Methods for Rubber Property - Adhesion to Flexible Substrate. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Life preservers furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Material. In instances where the drawings specified herein do not agree with the material or design specified in this specification, this specification shall govern.

3.3.1 Coated cloth and tape. The coated cloth used in the construction of the life preserver shall conform to type I of MIL-C-19002. The coated cloth tape used in the construction of the life preserver shall conform to type III of MIL-C-19002. The coated cloth and tape shall be of a quality that will withstand the performance requirements specified herein when fabricated into a life preserver.

3.3.2 Webbing. The dyed nylon webbing used in the life preserver harness shall conform to MIL-W-17337 and shall be of the following widths:

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Part	Width (inches)
Body harness webbing	1-1/2
Back harness webbing	1-1/2

3.3.3 Mechanical inflation system.

3.3.3.1 Holder assemblies. Each life preserver shall be provided with one holder assembly conforming to dimensions shown on Drawing S9300-1411524. The assembly shall consist of a holder constructed for use with the CO_2 cylinder conforming to MIL-C-16385, and shall have an external actuating lever with lanyard, and internal piercing pin, and a coin set screw. The holder shall be a high pressure aluminum die casting, and shall be anodized and treated with dichromate seal. The holder shall be fitted with an O-ring gasket as shown on Drawing S3306-1220373 and conforming to class B, no. 12, AN-6227 of MIL-P-5516. The configuration shall be such that the cylinder will be punctured and the piercing pin will be retracted with a travel of the actuating lever through an arc of not over 90 degrees. No side pull shall be required to puncture the cylinder and retract the pin. The discharge mechanism shall operate on the eccentric principle which will allow the piercing point to return to its original position after operation even when the lever is held in the tripped position. When not in use, there shall be provision for holding the actuating lever against the side of the holder by a length of 0.016 inch diameter aluminum safety wire. The lever shall be corrosion-resisting steel conforming to class FS310 of QQ-S-766. The lever shall be held in place with a staked corrosion-resisting steel pin conforming to class FS310 of QQ-S-763. The coin set screw shall be corrosion-resisting steel conforming to class FS310 of QQ-S-763. The holder threads shall be coated with a silicone grease after the holder has been assembled on the life preserver. There shall be a phosphor-bronze spring to ensure that the piercing pin separates from the cylinder immediately after puncturing to preclude any possibility of delay in inflation by the pin remaining in and sealing the hole. The piercing pin shall be hardened K-monel or KR-monel conforming to QQ-N-286. The piercing pin shall have a 60-degree point and shall be 5/64 inch diameter. The piercing pin shall be sealed so that there shall be no leakage of gas from the holder when the discharge lever is in any position. An O-ring may be provided on the shank of the piercing pin to accomplish this. The piercing pin in its normal position shall be at least 0.010 inch behind the cylinder seat to prevent accidental puncture. The holder shall be constructed to provide adequate clearance between the open end of the holder and the CO_2 cylinder so that in the event that the seating surface (where the puncturable end of the cylinder seats within the holder) becomes worn, the cylinder can still be screwed into the holder to a greater depth and onto its seat. The design of the entire inflation system shall be submitted to the Command or agency concerned for approval. Holders shall withstand a hydrostatic pressure of 3,500 pounds per square inch (lb/in^2).

3.3.3.2 Holder lanyard. The holder lanyard shall consist of a 1/8 inch diameter commercial braided nylon cord, a plastic button, and two black nickel-plated brass clips. When attached to the actuating lever, the cord shall be doubled back upon itself to form a loop. The two free ends of the cord shall be inserted through the holes in the bottom of the pull cord button and then tied in a sheet bend knot. A cord clip shall be attached to the cord adjacent to the pull cord button and adjacent to the discharge lever to hold the double cord

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together. The entire assembly of cord, clips, and buttons shall not exceed 2-1/2 inches in overall length. The discharge lever, cord, and button shall withstand a minimum pull of 100 pounds.

3.3.3.3 Manifold. The holder shall be securely attached to the life preserver by means of a brass manifold molded to a properly compounded neoprene base, which shall in turn be cemented to the inside surface of the life preserver fabric so that the holder projects on the outside. The brass manifold, including threads, shall be coated with electrodeposited nickel conforming to grade E, class 1 of QQ-N-290. The manifold and holder shall be equipped with flats to maintain the holder in fixed position. The manifold shall conform to piece no. 26 of Drawing S3306-1220373, and shall be properly constructed to fit the holder. The manifold shall be located and oriented as shown on Drawing S3306-1220373. The manifold shall be provided with a one-way check valve conforming to Drawing S3306-1183395. The holder and manifold shall be assembled as shown on Drawing S3306-1220373. Two synthetic rubber gaskets conforming to MIL-R-900 shall be provided, one for each side of the holder attachment. The hexagonal nut shall be nickel-plated brass.

3.3.4 Oral inflation valve. The oral inflation valve shall be in accordance with type II, class 1 of MIL-O-81375, except that the weight shall be 30.0 grams maximum, and the valve shall withstand without damage inflation by means of a blow gun having a discharge nozzle not exceeding 1/8 inch diameter when the gun is connected to a 225 lb/in^2 gauge pressure source.

3.3.5 Color. The color of the life preserver coated cloth and harness shall match the dry standard color chip conforming to DOD-E-699.

3.3.6 Thread. Thread used for sewing the harness shall conform to type II, size E of V-T-295, colored to match the harness webbing.

3.3.7 Harness hardware. Metal harness hardware shall conform to Drawing S3306-1220375, and shall be brass equal to or better than the requirements of QQ-B-613 and shall have a lusterless black finish conforming to MIL-F-495.

3.3.8 Adhesive. The adhesive used shall be polychloroprene conforming to class 1, 2, or 3 of MIL-A-5540, except the qualification tests are not applicable, the color shall acceptably match the life preserver coated cloth, and for the adhesion tests, the life preserver coated cloth shall be substituted for the cotton duck.

3.3.9 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.4 Construction. The construction details of the life preservers shall conform to Drawings S3306-1220372, S3306-1220373, S3306-1220374, S3306-1220375, and S9300-1411524.

3.4.1 Oral inflation. The oral inflation valve shall be installed on the outside of the life preserver, and shall communicate with the air compartment by means of a properly compounded neoprene connector 3/8 inch (minimum) inside diameter, 9/16 inch outside diameter. The construction of the oral inflation assembly shall be in accordance with Drawing S3306-1220373. The location of the oral inflation assembly shall be as shown on Drawing S3306-1220374. The valve shall be secured to the connector by a clamp built to withstand a load of 50 pounds.

3.4.2 Snap fasteners. The snap fasteners, piece no. 27, used for the CO_2 inflator cover shall be attached as shown in detail A of Drawing S3306-1220374, and shall be located as shown on Drawing S3306-1220372. Three sets of fasteners shall be furnished. Snap fasteners shall be in accordance with Drawing AN-227, 49B, 50B, 51B, and 52B, and shall have a black finish.

3.4.3 Inflation tube loop, and CO_2 cylinder loop. The inflation tube loop and CO_2 cylinder loop shall be as shown on Drawing S3306-1220374, and shall be made from two plies of coated cloth (see 3.3.1).

3.4.4 Cementing. In the construction of the seams, sealing area, and attachment of all component parts of the life preserver, either cementing or pressure-curing by heat shall be used. In all cementing operations, the surfaces to which the adhesive is to be applied shall be thoroughly washed, if necessary, with a solvent so that the dusting materials (zinc stearate or talc) and any other surface contaminant is completely removed. The surface shall be cleaned prior to cementing. Care shall be exercised to ensure that neither the coating nor the base undercloth is damaged, nor that the adhesion between both is impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive, and shall leave no residue. Cemented areas shall be rolled and stitched out, and all trapped air and wrinkles that may impair the function of the preserver shall be removed. Cements shall be controlled to ensure that old cement or cement which has partly set up is not used. A fresh batch of cement shall be mixed at least every 8 hours during the course of manufacture. Containers for cement shall be free from congealed cement before being refilled.

3.4.4.1 Cemented area. The cement on all outside cemented areas in way of taped seams and patches shall be applied to the area, and shall extend not less than 1/8 inch and not more than 3/8 inch beyond the edges of the tape or patch and parallel thereto.

3.4.4.2 Cementing seams and sealing areas. The construction of seams and sealing areas and the attachment of all component parts to the preserver shall be as specified in 3.4.4 so that the completed seams and sealed areas shall meet the optimum adhesive requirements as specified in MIL-A-5540.

3.4.5 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

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3.5 Marking. Identification marking shall be by the silk screen method in black letters and numerals 1/4 inch high. A marking patch, piece no. 25, shall be located on the outside back of the life preserver as shown on Drawing S3306-1220372. The date of manufacture and serial numbers may be rubber stamped.

3.6 Workmanship. Workmanship shall be in accordance with high-grade manufacturing practice covering this type of work. Because of the emergency use of this equipment, all details of workmanship shall be of the highest quality to ensure proper functioning of the preservers at all times.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.6).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. The inspection performed on the completed life preservers and component material shall consist of all the tests specified in this specification, including examination to determine conformance to the construction requirements. Samples of all materials used in the fabrication of the life preserver shall be examined and tested to ascertain their conformance to the applicable specifications or to the requirements for the material as specified herein.

4.3.1 Waiver of qualification. Qualification of life preservers under MIL-L-15581 constitutes qualification of life preservers under this specification.

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4.3.2 Samples for qualification inspection. The following samples shall be submitted for qualification inspection:

(a) Two complete life preservers.
(b) All coated fabric samples specified in MIL-C-19002.
(c) Uncoated cloth sample as specified in MIL-C-19377.
(d) Five yards of nylon webbing used in the life preserver.
(e) One cone (500 yards) of nylon sewing thread of each color used in the life preserver.
(f) Four sets of all hardware, including the oral and CO_2 inflation systems used in the life preservers.
(g) Five yards of safety wire.

4.3.2.1 Identification of qualification samples. Qualification inspection samples shall be plainly identified by securely attached durable tags marked with the following information:

(a) Samples for qualification inspection.
(b) Name of product (together with complete identifying data such as manufacturer's designation or number).
(c) Name of manufacturer.
(d) Submitted by (name) (date) for qualification inspection in accordance with MIL-L-16383 under authorization (reference authorizing letter).

4.4 First article inspection. First article inspection shall consist of the examination specified in 4.7 and the tests specified in 4.10.

4.4.1 First article samples. Prior to starting regular production, the contractor shall submit the following samples:

(a) Two complete life preservers.
(b) All coated fabric samples specified in MIL-C-19002.
(c) Uncoated cloth sample as specified in MIL-C-19377.
(d) Five yards of nylon webbing used in the life preserver.
(e) One cone (500 yards) of nylon sewing thread of each color used in the life preserver.
(f) Four sets of all hardware, including the oral and CO_2 inflation systems used in life preservers.
(g) Five yards of safety wire.

4.4.2 Forwarding of first article test samples. First article test samples shall be forwarded to the Commanding Officer and Director, U.S. Naval Applied Science Laboratory, Code 9380, Naval Base, Brooklyn, New York, 11251, plainly identified by securely attached durable tags marked with the following information:

(a) Samples for first article test.
(b) Name of product, together with complete identifying data, such as manufacturer's designation number.
(c) Name of manufacturer.
(d) Contract or order number.

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4.5 Quality control. The results of pressure, leakage, and buoyancy tests shall be identifiable by listing the assigned serial numbers. To ensure that the final product is sound and in conformance to the requirements of this specification, the contractor shall provide for an adequate control of quality of the coated fabric, the adhesive, and the attachments during the process of manufacture.

4.5.1 Quality control tests. The following tests shall be conducted on each life preserver by the contractor under the sureveillance of the inspector in the order listed. Life preservers not meeting the requirements specified herein shall be corrected prior to submitting the life preservers for acceptance, but under no circumstances shall an air retaining seam be reworked more than one time.

Test	Paragraph
Operation (CO-2)	4.10.1
Pressure (5 lb/in^2)	4.10.2
Leakage (2 lb/in^2)	4.10.3

The following additional tests shall be conducted by either the contractor or the subcontractor at the place of manufacture.

	Tests	Paragraph	Sampling plan
(a)	Leakage and operation (holder assembly)	4.10.5	100 percent
(b)	Oral inflation tube (porosity)	4.10.6.1	100 percent
(c)	Oral inflation tube (tensile)	4.10.6.2	Six samples from each 1000
(d)	Dimensional inspection (gauge) of holder assembly	4.10.7	100 percent
(e)	Oral inflation valve adhesion	4.10.6	Six samples from each 1000

4.5.2 Additional quality control test. In addition, the contractor shall prepare a minimum of ten adhesion test samples from each day's production, and shall subject the samples to the test specified in 4.10.8. The dimensions of the test specimen shall be in accordance with the requirements for strip specimens in accordance with the machine method of ASTM D 413.

4.6 Quality conformance inspection. Quality conformance tests shall be performed by or under the surveillance of the inspector on lots submitted for acceptance under the contract. Quality conformance tests shall be conducted after the life preservers have passed the examination specified in 4.7 and the tests specified in 4.5.1 and 4.5.2. The tests shall consist of the tests specified in 4.10.1 through 4.10.4. The buoyancy test (see 4.10.4) shall be conducted when the CO_2 cylinder is expended during the operation test (see 4.10.1).

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4.6.1 Lot. A lot shall consist of all life preservers submitted for inspection at one time.

4.6.2 Sampling for examination. A random sample of life preservers shall be selected from each lot in accordance with inspection level I of MIL-STD-105 for the examination specified in 4.7. The acceptance number shall be zero for critical defects. The acceptable quality level shall be 4.0 for major defects and 10.0 for minor defects per 100 units.

4.6.3 Sampling for tests. A random sample of life preservers shall be selected from each lot in accordance with the following inspection levels of MIL-STD-105 for the tests specified in 4.8:

Tests	Inspection level
Operation (CO_2) (see 4.10.1)	I
Pressure (5 lb/in^2) (see 4.10.2)	I
Leakage (2 lb/in^2) (see 4.10.3)	I
Buoyancy (see 4.10.4)	S-3

The acceptance number shall be zero for all tests. Rejected life preservers shall not be resubmitted.

4.7 Examination. Sample life preservers selected in accordance with 4.6.2 shall be thoroughly examined to determine compliance with the requirements of this specification and the applicable drawings. Examination shall be conducted as specified in table I.

TABLE I. Classification of defects in accordance with MIL-STD-105.

Categories	Defects
Critical:	
1	Buoyancy chamber material defective; evidence of splits, tears, patched holes and fraying.
2	Buoyancy chamber seams defective, incomplete, or not as specified; evidence of bubbles, improperly applied tape, incomplete cementing, or seams in buoyancy chamber not the specified type; width of seams less than the required minimum.
3	Manifold or oral inflation tube defective, not secure, securing patches not as required, or diameter of base not as specified.
4	Oral inflation valve defective or not secure in tube; tube evidences cracks, splits, bubbles, bulges, pin holes or other deleterious effects.
5	Holder assembly defective, not securely attached, or not as specified, lanyard not securely attached to actuating lever, lever holding pin not staked in place.
6	Failure of specified gauge to seat firmly against seating surface of holder, O-rings (for cylinder and piercing pin) missing, defective, or not as required (examined prior to assembly).
7	Gaskets (each side of holder attachment) missing, defective, or not as required.

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TABLE I. Classification of defects in accordance with MIL-STD-105. - Continued

Categories	Defects
Major:	
101	Loops for securing CO_2 cylinder or inflation tube defective or not securely attached as specified.
102	Holder and marking patch not secure, inflator cover not secured as required, or snap fasteners missing, defective, or not secured to patches as required.
103	Body and back harness webbing evidences tears, splits, fraying, or weak spots.
104	Harness stitching defective, incomplete, or not as specified.
105	Harness hardware missing, defective, or not secure; evidence of cracks, deep pits, thin spots, or sharp surfaces.
106	Holder actuating lever not safety wired.
107	Metal components (harness hardware, holder, lever, or manifold) not plated or finished as specified; plating or finish defective or incomplete.
108	Date of manufacture marking missing, illegible, or incorrect.
109	Coin set screw provided in holder for handling CO_2 cylinder missing or not staked.
110	Spare holder assembly not provided, incomplete, damaged, not as specified, or not wrapped as required.
111	Evidence of reworking air retaining seams more than the allowable number of times.
112	Holder threads not greased as required after assembly on life preserver.
Minor:	
201	Length of lanyard not as required.
202	Color of cloth, webbing, thread, and cement not as specified.
203	Marking, manufacturer's name, or trademark and identification missing, incorrect, illegible, not permanent, or not located as specified.
204	Packaging and packing not as specified.
205	Cement on fabric surface around accessory patches and tapes below the minimum limit of 1/8 inch and in excess of 1/2 inch.

4.7.1 Resubmitted lots. Resubmitted lots (allowed in connection with major and minor defects found in the examination) may be resubmitted for examination as specified in 4.6.2. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be selected, with the acceptance number remaining zero. The resubmitted lot shall be examined by the contractor under the surveillance of the inspector using tightened examination. Before a lot is resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct the defects found in the lot shall be furnished by the contractor to the inspector.

4.8 Comparison tests. NAVSEA may conduct at a Government laboratory, comparison tests on randon samples of life preservers produced under any contract.

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Comparison tests may include any or all of the tests of this specification. Acceptance and shipment of life preservers shall not be delayed pending completion of comparison tests. However, if any comparison test sample is rejected for noncompliance with the requirements of this specification, acceptance and shipment of any further life preservers shall be withheld until the contractor can demonstrate that he has corrected the conditions which led to the rejection of the comparison test sample. Samples selected for comparison tests shall be considered a part of the quantity to be delivered under contract and shall be returned with all applicable test reports, recommendations, and comments pertinent for monitoring production.

4.9 Test conditions.

4.9.1 Atmospheric conditions. Unless otherwise specified in the contract or order (see 6.2), all tests required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury (Hg), a temperature of 77 ± 18 degrees Fahrenheit (°F), and a relative humidity of 80 percent or less. If the final values of the ambient temperature or barometric pressure, at the end of the 24-hour air pressure tests are different from the initial values recorded at the start of the test, the following corrections shall be made to the final pressure readings in lb/in^2:

(a) Temperature corrections for each degree rise in temperature, 0.031 lb/in^2 shall be subtracted from the final pressure reading. For each degree drop in temperature, 0.031 lb/in^2 shall be added to the final pressure reading.
(b) Barometric pressure correction - For each 0.1 inch of Hg rise in barometric pressure, 0.049 lb/in^2 shall be added to the final temperature corrected pressure reading. For each 0.1 inch of Hg drop in barometric pressure, 0.049 shall be subtracted from the final temperature corrected pressure reading.
(c) Pressure shall be measured by means of a Hg manometer and then converted to lb/in^2 by multiplying inches of Hg by 0.49.

4.9.2 Test CO_2 cylinder requirements. CO_2 cylinders used in conducting the tests specified in 4.10.1 and 4.10.4 shall be in accordance with MIL-C-25369, except that the gas charge shall be not less than 17 grams nor more than 24 grams of CO_2. The cylinder shall be marked with a permanent type ink as follows: "For Test Purposes Only" and the inclusion of a nonapplicable contract number or the omission of the contract number is acceptable.

4.10 Test methods.

4.10.1 Operation (CO_2). The operation of the preserver shall be tested by releasing a CO_2 cylinder into the life preserver in the normal manner. The life preserver shall inflate instantly.

4.10.2 Pressure (5 lb/in2). The life preserver shall be inflated with air through the oral inflation valve to a pressure of 5 lb/in^2 and shall be allowed to stand for a minimum of 5 minutes. The pressure drop shall be not greater than 0.4 lb/in^2. The life preserver shall be examined while held under pressure.

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There shall be no evidence of failure of the materials in any respect. All seams and sealed areas shall remain perfectly intact and shall show no signs of separation.

4.10.3 Leakage (2 lb/in2). The life preserver shall be inflated with air through the oral inflation valve to a pressure of 2 lb/in^2 and shall be allowed to stand for a minimum of 15 minutes. The pressure shall then be checked and readjusted if necessary to the original pressure of 2 lb/in^2. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 24 hours after the readjustment period, the pressure shall be measured and corrected for any change in temperature or barometric pressure (see 4.9.1). The corrected pressure shall be not less than 1.5 lb/in^2. The oral inflation valve shall be examined when the life preserver is being deflated to determine that the valve works without difficulty and shuts off securely.

4.10.4 Buoyancy. Immediately upon completion of the operation (CO_2) test (see 4.10.1), with the oral inflation valve locked shut, the life preserver shall be attached to a steel weight weighing 19 pounds when submerged in fresh, clean water. The life preserver shall support this 19-pound weight without sinking.

4.10.5 Leakage and operation (holder assembly). Holders shall be tested for leakage underwater with air at a pressure of 40 lb/in^2 with the actuating lever in the actuated position. (The actuated position is considered to be the position of the lever after the pin has punctured the cylinder and has completely retracted. Cylinders are not required to determine the actuated position.) Holders shall withstand a hydrostatic pressure of 3,500 lb/in^2. Holders shall also be tested for leakage underwater with the actuating lever slowly carried through one complete cycle.

4.10.6 Oral inflation tube. Before the oral inflation valve is clamped in the oral inflation tube, the tests specified in 4.10.6.1 and 4.10.6.2 shall be conducted.

4.10.6.1 Porosity. The oral inflation tube shall be subjected to an air pressure of 25 lb/in^2 for a minimum of 10 seconds while submerged underwater. There shall be no evidence of air leakage due to porosity. There shall be no evidence of weakening of the tube as revealed by excessive deformation in any portion of the tube.

4.10.6.2 Tensile. The oral inflation tube base shall be folded longitudinally so that the rounded edges coincide, and then placed in the lower jaws of a suitable test machine. The tube portion shall be placed in the upper jaws. The rate of load application shall be 12 $\pm$ 1/2 inches per minute. The tensile strength shall be not less than 75 pounds.

4.10.7 Dimensional inspection (gauge) of holder assembly. Holder assembly (complete with O-ring) shall be dimensionally checked with a plug gauge conforming to the dimensions shown on Drawing S9300-1411524. Failure of the plug gauge to seat firmly against seating surface (where the puncturable end of the cylinder seats within the holder) shall be cause for rejection.

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4.10.8 Adhesion. The test for adhesion shall be conducted in accordance with the machine method of ASTM D 413. The adhesive strength of the cemented samples shall be not less than 7.5 pounds per inch of width for samples joining the coated sides of the finished fabric and not less than 5.0 pounds per inch of width for seams joining coated side to uncoated side of the finished fabric.

4.10.9 Salt spray. A 100-hour salt spray test shall be conducted on the CO_2 inflation devices, oral inflation valve, and all metal parts in accordance with method 811.1 of FED-STD-151. The metal fittings shall show no appreciable corrosion or other damage at the end of this test.

4.10.10 Oral inflation valve adhesion. This test shall be conducted after the oral inflation valve is clamped within the oral inflation tube. The oral inflation tube base shall be folded longitudinally so that the rounded edges coincide, and then placed in the lower jaws of a suitable machine. The oral inflation valve shall be placed in the upper jaws. The rate of load application shall be 12 $\pm$ 1/2 inches per minute. The rate of load to separate the oral inflation valve from the oral inflation tube shall be not less than 50 pounds.

4.11 Possible test failures. Possible test failures shall be determined as follows:

(a) Operation of assembled life preserver:
  (1) Failure of life preserver to inflate when lanyard is pulled; evidence of leaks through fabric, seams, holder or piercing pin O-rings, or manifold check valve after inflation.
  (2) Life preserver cannot be readily inflated through the oral inflation valve or evidence of leakage back through the valve after inflation.

(b) Test of operation or function:
  (1) Displacement of life preserver not within the required maximum and minimum values when inflated and measured as specified.

(c) Test of leakage:
  (1) Displacement of life preserver less than the minimum permissible after it has been inflated and then allowed to stand for the required length of time.

(d) Buoyancy:
  (1) Failure of the inflated life preserver to support, without sinking, the specified weight.

(e) Pull test:
  (1) Discharge lever, cord, and button fail to withstand the required minimum pull.

4.12 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging, packing, and marking. Life preservers shall be packaged level A or C; packed and marked level A, B or C as specified (see 6.2) in accordance with MIL-L-3454.

6. NOTES

6.1 Intended use. The life preservers covered by this specification are intended for use as emergency equipment by underwater demolition teams.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) If atmospheric test conditions are other than specified (see 4.9.1).
(c) Level of packaging, packing and marking (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-16383 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of units specified in 4.4.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Subject term (key word) listing.

Cloth, coated
Holder assembly
Inflation valve
Lanyard
Manifold
Nylon
Webbing

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4420-N068)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-16383H(SH)	LIFE PRESERVER, CO_2 INFLATABLE, UNDERWATER DEMOLITION TEAM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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